Soccer Training App - Complete Product Knowledge

Product Overview

AI-powered soccer training app that analyzes drill performance using computer vision to detect ball control and provide FIFA-style ratings and feedback.

Core Value Proposition: "See your child's soccer development, not just game highlights" - Multi-year development tracking with FIFA-style ratings.

Business Model & Market

- **Target Market**: 4M youth soccer players in US, parents spending $3,000+ annually per child
- **Market Size**: $10.2B youth sports software market growing 11.5% annually
- **Revenue Model**:
  - Family: $39/month (development tracking, progress reports)
  - Team: $99/month (coach tools, parent communication)
  - Club: $299/month (multi-team management, recruiting integration)
- **Target**: 100K families × $450 average = $45M ARR within 5 years

Technical Architecture

Frontend Stack

- **Framework**: React Native with Expo SDK 53
- **Navigation**: React Navigation v7
- **Camera**: react-native-vision-camera (replaced expo-camera)
- **State Management**: React Context + useReducer
- **Styling**: NativeWind (Tailwind for React Native)
- **Build System**: EAS Build (Expo Application Services)

Backend Stack

- **Framework**: Python FastAPI
- **Computer Vision**: YOLO v8 + OpenCV + MediaPipe
- **Database**: Supabase (PostgreSQL)
- **Authentication**: Supabase Auth

- **Server**: DigitalOcean Droplet (Ubuntu, IP: 147.182.224.87)
- **Video Processing**: FFmpeg for compression

Core AI Pipeline

1. **Ball Detection**: Custom YOLO v2 model (76.5% mAP, 13.8% detection rate)
2. **Pose Detection**: MediaPipe for player foot tracking
3. **Touch Logic**: Ball proximity to feet + velocity threshold + 500ms debounce
4. **Fallback Systems**: HSV color detection + Hough circle detection

Services & Platforms

Development & Deployment

1. **Claude Code** - AI-assisted development environment
2. **GitHub** - Code repository (soccer-training-app)
3. **Expo/EAS** - App building & deployment platform
4. **Apple Developer** - App Store Connect & TestFlight
5. **DigitalOcean** - Backend server hosting

Data & Analytics

6. **Supabase** - Database, auth, and real-time features
7. **YOLO (Ultralytics)** - Computer vision model
8. **OpenCV** - Video processing library
9. **MediaPipe** - Pose detection
10. **FFmpeg** - Video compression

Current Features

Drill Analysis System (8 Supported Drills)

1. **Juggling** - Keep ball in air (20+ touches/min for 30s+)
2. **Bell Touches** - Alternating foot taps (18-24 touches in 30s)
3. **Inside-Outside** - Same foot alternating touches (12-18 per foot)
4. **Sole Rolls** - Rolling with bottom of foot (8-14 in 20-30s)
5. **Outside Foot Push** - Outside of foot only (15-22 in 30s)
6. **V Cuts** - Pull-push patterns (6-10 per foot in 20-30s)
7. **Croquetas** - Side-to-side cuts (8-15 in 15-30s)
8. **Triangles** - Triangle movement patterns (4-8 patterns in 20-30s)

FIFA-Style Rating System

- **Overall Rating**: 45-85 based on age (like FIFA Career Mode)
- **Skill Categories**: Pace, Ball Control, Shooting, Dribbling, Passing, Defending, Physical, Mental
- **Age-Adjusted Benchmarks**:
  - Age 8: Average 45 OVR, Elite 60 OVR
  - Age 12: Average 55 OVR, Elite 70 OVR
  - Age 16: Average 65 OVR, Elite 80 OVR

API Architecture

Drill Management Endpoints

- `GET /drill/available` - List all 8 drills with descriptions
- `GET /drill/types` - Simple array of drill IDs
- `GET /drill/info/{drill_type}` - Detailed drill information
- `GET /drill/benchmark/{drill_type}` - Success criteria

Video Analysis Endpoints

- `POST /drill/analyze` - Upload video + drill type selection
- `POST /drill/analyze/{video_id}` - Analyze existing video for new drill
- `GET /drill/status/{analysis_id}` - Processing status
- `GET /drill/results/{analysis_id}` - Complete analysis results

Legacy Endpoints (Keep-ups focus)

- `POST /upload` - Upload video for analysis
- `GET /status/{video_id}` - Check processing status
- `GET /results/{video_id}` - Get analysis results
- `GET /videos` - List all processed videos
- `DELETE /video/{video_id}` - Clean up old videos

Database Schema (Supabase)

Core Tables

```
-- Videos table
videos {
  id: uuid (primary key)
  filename: text
  file_path: text
```

```
  status: text (uploaded/processing/completed/error)
  results_path: text
  total_touches: integer
  confidence_score: float
  drill_type: text
  created_at: timestamp
  updated_at: timestamp
}

-- User profiles (planned)
profiles {
  id: uuid (references auth.users)
  age: integer
  overall_rating: integer
  skill_ratings: jsonb
  created_at: timestamp
}
```

File Structure

Backend Structure

```
/backend/
├── main.py                 # FastAPI application server
├── video_processor.py      # Core CV analysis engine
├── database.py             # Supabase database interface
├── drill_analyzer.py       # Unified drill analysis framework
├── drill_api.py            # Drill-specific API endpoints
├── unified_processor.py    # Video processing orchestrator
├── models/                 # AI model files
│   ├── soccer_ball_detector/     # Original YouTube-trained model
│   └── real_detector_v2/         # Improved model (13.8% detection)
├── uploads/
│   ├── raw/              # Original uploaded videos
│   ├── processed/        # JSON analysis results
│   └── frames/           # Debug screenshots
└── training_data/        # AI training pipeline
```

Frontend Structure

```
/frontend/
├── App.js              # Main app component
├── src/
```

```
│   ├── components/      # Reusable UI components
│   ├── screens/         # Main app screens
│   ├── services/        # API calls and data management
│   ├── utils/           # Helper functions
│   └── contexts/        # React contexts for state
├── assets/              # Images, fonts, etc.
└── app.json             # Expo configuration
```

Current Performance Metrics

- **Ball Detection Accuracy**: 76-88% (target: 90%+)
- **Processing Time**: ~45 seconds per 30-second video
- **Model Performance**: 13.8% detection rate (5.75x improvement from baseline)
- **Success Rate**: 60-85% touch detection accuracy

Development Workflow

Local Development

```
# Backend setup
cd backend
python -m venv venv
source venv/bin/activate  # or venv\Scripts\activate on Windows
pip install -r requirements.txt
uvicorn main:app --reload --host 0.0.0.0 --port 8000

# Frontend setup
cd frontend
npm install
npx expo start
```

Deployment Process

```
# Backend deployment (DigitalOcean)
ssh root@147.182.224.87
cd /root/soccer-app/backend
git pull origin main
sudo systemctl restart soccer-app

# Frontend deployment (EAS Build)
eas build --platform ios --profile production
eas submit --platform ios
```

Environment Variables

Backend (.env)

SUPABASE_URL=your_supabase_url
SUPABASE_ANON_KEY=your_anon_key
SUPABASE_SERVICE_KEY=your_service_key
UPLOAD_DIR=/uploads
MAX_FILE_SIZE=100MB
PROCESSING_TIMEOUT=300
MODEL_PATH=training_data/experiments/real_detector_v2/weights/best.pt

Frontend (.env)

EXPO_PUBLIC_API_URL=https://147.182.224.87:8000
EXPO_PUBLIC_SUPABASE_URL=your_supabase_url
EXPO_PUBLIC_SUPABASE_ANON_KEY=your_anon_key

Key Dependencies

Backend Requirements

fastapi==0.104.1
uvicorn==0.24.0
python-multipart==0.0.6
opencv-python==4.8.1.78
mediapipe==0.10.7
ultralytics==8.0.196
numpy==1.24.3
Pillow==10.0.1
supabase==1.2.0
python-dotenv==1.0.0

Frontend Dependencies

```
{
  "expo": "~53.0.0",
  "react-native": "0.76.0",
  "react-navigation": "^7.0.0",
  "react-native-vision-camera": "^4.0.0",
```

```
  "nativewind": "^2.0.11",
  "expo-av": "~15.0.0"
}
```

Known Issues & Solutions

Current Challenges

1. **Motion blur during ball-foot contact** - Planning ByteTrack integration
2. **Low-confidence detections being discarded** - YOLO v8 upgrade planned
3. **Frame-to-frame tracking inconsistency** - Kalman filter implementation needed
4. **Video quality variations** - Automatic enhancement planned

Implemented Solutions

1. **Camera replacement**: Switched from expo-camera to react-native-vision-camera for better performance
2. **Smart frame sampling**: Process frames near feet instead of all frames
3. **Multi-method detection**: YOLO + HSV color + Hough circles as fallbacks
4. **Confidence thresholds**: 0.05 threshold to catch more detections

Monitoring & Maintenance

Daily Checks (5 minutes)

- [ ] DigitalOcean server uptime (test app functionality)
- [ ] Expo dashboard for build status and crash reports
- [ ] TestFlight for new beta tester feedback

Weekly Checks (30 minutes)

- [ ] Supabase database usage and storage limits
- [ ] GitHub repository activity and issues
- [ ] Apple Developer Console for app review status
- [ ] Model performance metrics and accuracy trends

Monthly Reviews (2 hours)

- [ ] DigitalOcean billing and resource usage
- [ ] Expo build usage and plan limits
- [ ] Training data collection and model retraining
- [ ] User feedback analysis and feature prioritization

Competitive Advantages

Technical Moat

- **AI Development Speed**: Ship features weekly vs competitors' quarterly cycles
- **Proprietary Training Data**: Every user video = new training examples
- **Multi-drill Architecture**: Same pipeline handles all skill types
- **Self-Improving System**: User corrections = labeled improvement data

Market Position

- **First FIFA-style rating system** for youth soccer development
- **Privacy-first approach**: No video storage by default (COPPA compliant)
- **Comprehensive platform** vs single-purpose apps
- **Backyard-friendly focus** vs field-based solutions

Roadmap & Next Steps

Phase 1 (Complete): MVP Keep-ups Detection

- ✅ Basic ball detection and touch counting
- ✅ Web app with video upload
- ✅ 76-88% accuracy achieved

Phase 2 (In Progress): Multi-Drill Platform

- ✅ 8 drill types supported in backend
- 🔄 Frontend drill selection interface
- 🔄 FIFA-style rating system implementation
- 🔄 ByteTrack + YOLO v8 upgrade for 90%+ accuracy

Phase 3 (Planned): Advanced Features

- [ ] Multi-year development tracking
- [ ] Parent dashboard with progress insights
- [ ] Coach integration and team features
- [ ] Social features and peer comparisons

Phase 4 (Future): Scale & Expansion

- [ ] Multi-sport expansion
- [ ] Recruiting integration
- [ ] Advanced game analysis

- [ ] Team tactical analysis

Cost Structure

Monthly Operating Costs

- **Supabase**: Free tier (planning upgrade to $25/month)
- **DigitalOcean**: $6-12/month (1 CPU, 1GB RAM droplet)
- **Expo**: Paid plan for unlimited builds (~$29/month)
- **Apple Developer**: $99/year
- **Total**: ~$40-50/month operating costs

Development Costs Saved

- **Traditional Development**: 5 months, $150,000+ in developer costs
- **With Claude Code**: 5-6 weeks, $20,000-25,000 investment
- **Savings**: 80% faster, 85% cheaper development

Success Metrics

Technical KPIs

- **Detection Accuracy**: Currently 76-88%, target 90%+
- **Processing Time**: Currently ~45s, target <30s
- **Uptime**: Target 99.9% server availability
- **User Experience**: <3 minute end-to-end analysis time

Business KPIs

- **User Retention**: Multi-year tracking creates switching costs
- **Engagement**: Weekly drill completion rates
- **Accuracy Improvement**: Self-improving model performance
- **Revenue**: Path to $45M ARR with 100K families

This comprehensive knowledge base captures your complete soccer training platform - from technical architecture to business strategy. You can use this for Claude Desktop, team onboarding, or investor presentations!